Contact

+91 90000 22052 (Mobile)
nisha.iyer@example.com

www.linkedin.com/in/nisha-iyer-embedded-sample (LinkedIn)

Top Skills

Embedded C

C++

FreeRTOS

Languages

English (Professional Working)

Tamil (Native or Bilingual)

Hindi (Conversational)

Nisha Iyer

Embedded Systems Student | Firmware, IoT, Edge Sensing, Real-Time Systems | VIT Vellore

Chennai, Tamil Nadu, India

Summary

I am an electronics and communication engineering student focused on embedded firmware, low-power sensing, and dependable device-to-cloud systems. I have worked with STM32 and ESP32 platforms, FreeRTOS task design, communication buses, sensor calibration, and automated hardware-in-the-loop testing. I enjoy debugging across boundaries, from electrical signals and timing constraints to firmware state machines and cloud telemetry. I am seeking embedded software, firmware, or IoT internships where I can contribute to products that need careful resource use and reliable field behavior.

Experience

Asterion Mobility Labs

Embedded Software Intern

May 2026 - July 2026

Chennai, India

Implemented FreeRTOS tasks for battery sensing, motor-temperature monitoring, diagnostics, and CAN message handling on an STM32-based controller. Reduced telemetry packet loss from 4.7% to 0.6% by redesigning interrupt-to-task queues and adding bounded retry logic. Created Python hardware-in-the-loop tests for 38 fault conditions, including sensor disconnects, stale values, bus congestion, and brownout recovery. Documented timing budgets and stack usage for six tasks, enabling safer priority and memory configuration before road testing.

VIT Embedded Intelligence Lab

Student Research Assistant

September 2025 - Present

Vellore, India

Designed a low-power vibration-sensing node for predictive maintenance experiments using accelerometer sampling, local feature extraction, and LoRaWAN uplinks. Optimized duty cycling and fixed-point signal processing to extend estimated battery life from 11 weeks to 19 weeks. Collected labeled motor-fault datasets and evaluated threshold, random forest, and one-class anomaly approaches for edge deployment.

Synthetic QA LinkedIn-style profile - all details fictional

Core Tools

Embedded C

STM32

CAN

STM32CubeIDE

Interrupts

Portfolio

github.com/nishaiyer-embedded

nishaiyer.example.com

Focus Areas

Production systems

Testing and quality

Clear documentation

Measurable outcomes

Experience

CircuitSpring Academy

Firmware Trainee

December 2024 - March 2025

Remote

Completed project-based training in GPIO, timers, ADC, PWM, UART, SPI, I2C, debouncing, bootloaders, and defensive C programming. Built reusable peripheral drivers and unit-tested protocol parsing on the host before flashing target hardware.

Projects

Edge Predictive Maintenance Node

STM32, FreeRTOS, C, CMSIS-DSP, LoRaWAN

October 2025 - Present

Built a sensor node that samples three-axis vibration, computes RMS, kurtosis, crest factor, and band-energy features, and transmits compact health summaries. Implemented ring buffers and double-buffered DMA to maintain deterministic acquisition while communication tasks were active. Added remote configuration with versioned settings, CRC validation, and rollback to last-known-good parameters.

Low-Power Posture Wearable

ESP32, C++, BLE, IMU, Flutter

June 2025 - September 2025

Developed a wearable prototype that classifies sustained posture states and sends configurable feedback over Bluetooth Low Energy. Reduced average current draw by 37% using event-driven sampling, connection interval tuning, and deep-sleep transitions.

CAN Bus Diagnostic Analyzer

C++, Qt, SocketCAN, Python

February 2025 - May 2025

Created a desktop tool for filtering, decoding, replaying, and exporting CAN frames with message-rate and timeout visualization. Added DBC-based signal decoding and scripts that generated repeatable fault-injection traffic.

Smart Irrigation Controller

ESP32, MQTT, C, Node-RED, InfluxDB

August 2024 - December 2024

Automated zone watering using soil moisture, rainfall, reservoir level, and configurable quiet hours. Implemented local fallback rules so schedules continued during broker or Wi-Fi outages.

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Education

Vellore Institute of Technology

Bachelor of Technology - BTech, Electronics and Communication Engineering

Credentials

Arm Education: Embedded Systems Essentials - 2026

NPTEL Introduction to Internet of Things - 2025

Honors

VIT Research Seed Grant - Student Contributor, 2026

IEEE Hardware Hackathon Runner-Up, 2025

Profile Note

This is a synthetic profile created for document-processing and token-tuning QA.

Education

Vellore Institute of Technology
Bachelor of Technology - BTech, Electronics and Communication Engineering
July 2023 - May 2027

Relevant coursework: Embedded Systems, Digital Signal Processing, Microcontrollers, Computer Architecture, Communication Systems, Control Systems, Data Structures, IoT Security

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Licenses & Certifications

Arm Education: Embedded Systems Essentials - 2026
Issued for skills validation in this synthetic QA profile.

NPTEL Introduction to Internet of Things - 2025
Issued for skills validation in this synthetic QA profile.

Volunteer Experience

VIT Robotics Club
VIT Robotics Club - firmware lead for a six-member autonomous rover team.

IEEE Student Branch
IEEE Student Branch - conducted beginner workshops on microcontrollers, sensor interfaces, and debugging with logic analyzers.

Women in Hardware mentorship circle
Women in Hardware mentorship circle - paired with first-year students for semester-long build projects.

Skills

Programming
Embedded C, C++, Python, MATLAB, Bash

Platforms
STM32, ESP32, ARM Cortex-M, FreeRTOS, Arduino, Raspberry Pi

Interfaces
CAN, UART, SPI, I2C, BLE, MQTT, LoRaWAN

Tools
STM32CubeIDE, PlatformIO, Git, CMake, GDB, logic analyzer, oscilloscope

Concepts
Interrupts, DMA, fixed-point DSP, low-power design, bootloaders, HIL testing

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